

PROJECTS

Requirements:

- Group of 3 students. Each group picks one project.
- Survey and analyse the collected information
- Use **Extended ER (EER) features** (specialization/generalization) from Chapter 4
- Use the report structure in the provided template.
- Use industrial documentation standards and frameworks.
- Report weekly.

1. Smart City EV Charging & Parking Management

- **The Scenario:** A system to manage urban electric vehicle charging stations, parking slot bookings, and dynamic pricing based on peak hours.
- **DB Challenge:** Handling temporal data (reservations), calculating durations, and tracking varying rates per kilowatt-hour.

2. On-Demand Streaming Platform (Mini-Netflix)

- **The Scenario:** Managing user profiles, content categorization (movies vs. TV series episodes), watch histories, and subscription tiers.
- **DB Challenge:** Implementing EER hierarchies for Content (Movie vs. Episode), and managing many-to-many relationships for user watchlist histories.

3. Freelance Marketplace Ecosystem (Mini-Upwork)

- **The Scenario:** Connecting clients with freelancers, posting jobs, managing project milestones, contracts, and payment releases.
- **DB Challenge:** Handling multi-party transactions, milestone tracking, and user-rating aggregation tables.

4. University Research & Lab Asset Tracker

- **The Scenario:** Tracking university research grants, laboratory equipment, chemical inventories, and student/faculty access logs.
- **DB Challenge:** Enforcing strict integrity constraints (e.g., hazardous materials only accessible by certified personnel) via SQL Triggers.

5. Healthcare Clinic & Telemedicine Portal

- **The Scenario:** Coordinating patient appointments, medical histories, doctor schedules, and digital prescriptions.

- **DB Challenge:** Maintaining absolute data integrity and designing clean specialization for medical staff (General Practitioners vs. Specialists).

6. Supply Chain & Warehouse Logistics Network

- **The Scenario:** Tracking inventory items across multiple geographical warehouses, supplier orders, and shipment tracking.
- **DB Challenge:** Recursive relationships (components made of other sub-components) and managing inventory quantity triggers (restock alerts).

7. Food Delivery Aggregator (Mini-DoorDash)

- **The Scenario:** Connecting restaurants, customers, and delivery drivers. Tracking orders from placement, cooking, to real-time delivery status.
- **DB Challenge:** Complex multi-way joins (Customer-Restaurant-Driver-Order) and handling geometric/location coordinates if using spatial extensions.

8. Digital Library & Open-Access Repository

- **The Scenario:** Managing traditional physical book loans alongside digital journal access, open-access licenses, and fine collection.
- **DB Challenge:** Complex business rules for borrowing limits, overlapping entity types (e.g., an author who is also a university member).

9. Gym Membership & Personal Training Planner

- **The Scenario:** Tracking members, class schedules, personal trainer bookings, and fitness facility maintenance logs.
- **DB Challenge:** Avoiding scheduling conflicts via database constraints and handling recurring subscription billing logs.

10. Real Estate Property & Rental Ledger

- **The Scenario:** A platform for listing properties for sale or rent, tracking agent allocations, tenant leases, and maintenance requests.
- **DB Challenge:** Transitioning properties between "Available", "Leased", and "Under Maintenance" states while preserving historical financial logs.